FYP DEFENSE • 10-MINUTE PRESENTATION SCRIPT

MARP — Multi-Agent Market Research Platform

Next-Generation Out-Of-Home & Retail Field Research Platform

Duration: 10:00 Min + 5 Min Q&A
Frontend: marp-platform.vercel.app
Backend: marp-production.up.railway.app

1 Aroon Kumar
Seat No: 2312466

Module 01: User & Workspace Mgmt
Multi-tenant isolation, RBAC (5 roles), IMEI device security, audit logs.

2 Mudasir Ali
Seat No: 2312465

Module 02: Questionnaire Logic
10 field question types, branching engine, Urdu RTL, immutable versions.

3 Abdullah Zafar Khan
Seat No: 2312467

Module 03: Campaigns & Assets
5-tier location tree, billboard registry, competitor intel, evidence geofence audit.

10-Minute Defense Meeting Schedule

Exact 600-Second Allocation

ELAPSED TIME	SPEAKER	SECTION / KEY DISCUSSION	VISUAL / LIVE DEMONSTRATION
0:00 – 1:15	Aroon Kumar	Opening hook, Pakistan market fraud problem & annual Rs 35B spend.	Slide 1 (Title) & Slide 2 (Problem Statement)
1:15 – 3:15	Aroon Kumar	System Architecture & Module 01 (Multi-Tenancy, RBAC, Hardware Security).	Slide 3 & Slide 5 + Live Admin Login (/users, /settings)
3:15 – 4:45	Mudasir Ali	Module 02 Overview, 10 Question Types, Visual Form Designer.	Slide 6 + Live Questionnaire Builder (/questionnaires)
4:45 – 6:30	Mudasir Ali	Deterministic Branching Engine, Urdu RTL Toggle, Immutable Snapshots.	Live Branching Demo ('Poor' condition) + Urdu Preview
6:30 – 8:00	Abdullah Zafar	Module 03 Overview, 5-Tier Geo Tree, Billboard Media Registry.	Slide 7 + Location Hierarchy & Billboards (/campaigns)
8:00 – 9:15	Abdullah Zafar	Competitor Intelligence, Surveyor Dispatch & Geofence Fraud Audit.	Competitor Comparison + Evidence Rejection (340m flag)
9:15 – 10:00	Abdullah Zafar	Master 3-Circle ERD (28 Tables), Foreign Key Conduits & Conclusion.	Slide 4 (3-Circle ERD) & Slide 10 (Q&A Transition)

🎙️ PART 1: Aroon Kumar (Seat No: 2312466) — Lead Architect

`0:00 - 3:15 (3 Min 15 Sec)`

Visual Action: Display Slide 1 (Title Slide), then advance to Slide 2 (The Pakistan Problem).

> *"Respected supervisor, honorable committee members, and evaluators — Good afternoon.*
>
> *My name is **Aroon Kumar**, and alongside my project partners **Mudasir Ali** and **Abdullah Zafar Khan**, we are proud to present our Final Year Project: **MARP — Multi-Agent Market Research Platform**.*
>
> *In Pakistan today, corporate brands in telecommunications, FMCG, and banking spend over **35 billion rupees annually** on roadside billboards, digital SMDs, and outdoor media. Yet, almost every CMO and Brand Manager faces the exact same unresolved dilemma: **How do we reliably prove our advertisements are actually up, illuminated at night, and structurally undamaged?***
>
> *Traditional research still depends on clipboard paper surveys and informal WhatsApp photos. Surveyors forge data from home, paper records get lost, and enterprises have zero forensic proof of ground truth. MARP is engineered to eliminate this fraud entirely."*

Visual Action: Advance to Slide 3 (System Architecture), then open Live Browser at `/login` .

> *"To solve this, we did not build a superficial prototype. We engineered a **production-grade, full-stack enterprise web platform**:*
>
> • **Frontend:** Next.js 14 App Router, React 18, and custom Tailwind CSS tokens forming a sleek, ivory-and-slate design language responsive from 4K down to mobile.
> • **Backend:** Python FastAPI modular monolith with Pydantic v2 schemas and strict asynchronous endpoints hosted on Railway.
> • **Database:** PostgreSQL 16 managed via SQLAlchemy 2.0 and Alembic migrations, housing **28 normalized relational tables** with strict multi-tenant scoping.
>
> *(Log in as `admin@marp.io` and navigate to `/users`)*
>
> *My core responsibility was **Module 01: User Management, Workspace Isolation & Device Governance**. As you can observe on the live screen:*
>
> 1. **Multi-Tenant Workspace Isolation:** Every record is strictly partitioned by `company_id`. No client can ever access another tenant's proprietary research.
> 2. **Role-Based Access Control (RBAC):** We implemented five enforced roles: *Company Administrator, Research Manager, Analyst, Field Staff,* and *Evidence Reviewer*.
> 3. **Hardware IMEI Device Governance:** Under the Registered Devices tab, surveyor phones are authenticated via hardware identifiers. We can approve verified phones or instantly revoke compromised devices.
> 4. **Tamper-Evident Forensic Audit Trail:** Every privileged action — logins, role promotions, device approvals — is permanently recorded with actor identity, timestamp, IP, and JSON audit metadata.
>
> *With security and user governance firmly established, I will now hand over to **Mudasir Ali** to present **Module 02: Questionnaire Logic & Dynamic Builder**."*

🎙️ PART 2: Mudasir Ali (Seat No: 2312465) — Questionnaire Engine `3:15 - 6:30 (3 Min 15 Sec)`

Visual Action: **Advance to Slide 6 (Module 02), navigate live screen to `/questionnaires` .**

> *"Thank you, Aroon. Respected evaluators, my name is **Mudasir Ali**, and I spearheaded **Module 02: Questionnaire Management & Dynamic Logic Engine**.*
>
> *Market research demands versatile survey instruments. In our **Questionnaire Library**, surveys are categorized into four distinct research streams: **Billboard Audits, Retail Stock Checks, Competitor Studies,** and **Consumer Sampling**.*
>
> *(Click into 'National OOH Billboard Visibility Audit' → Visual Builder)*
>
> *In our Visual Questionnaire Builder, managers can assemble question sequences supporting **10 distinct field types**: short answers, paragraphs, single-choice radios, multi-select checkboxes, rating scales (1 to 5), dropdowns, photographic uploads, precision GPS pinpoints, and timestamps."*

Visual Action: **Switch to 'Conditional Branching' tab, then open 'Interactive Preview'.**

> *"A fundamental requirement of our project was that forms must not be static — they must execute **deterministic conditional branching logic**.*
>
> *Observe our rule engine: In Question 1, when evaluating billboard physical condition:*
>
> • If the surveyor selects *'Excellent'*, *'Good'*, or *'Average'*, the survey proceeds normally.
> • But if they select **'Poor'**, our branching engine dynamically intervenes and mandates **Question 3: 'Upload Close-up Photograph of Structural Damage or Torn Skin'**.
>
> *(In the Interactive Preview, click 'Poor' radio button → Question 3 immediately appears)*
>
> *Notice how Question 3 instantly materializes on the preview screen!*
>
> *Furthermore, field surveyors across Pakistan are bilingual. Watch what happens when I toggle this language switch:*
>
> *(Click 'اردو' language toggle)*
>
> *The entire questionnaire instantly re-renders in **Urdu with native Right-to-Left (RTL) Nastaliq alignment**, translating questions, options, and validation warnings seamlessly.*
>
> *Finally, to protect data integrity, whenever a questionnaire is published, our backend seals an **immutable JSON schema snapshot** as Version 1.0. Future revisions automatically fork into drafts without corrupting active field collection. We also integrated respondent eligibility quotas and SMS OTP verification.*
>
> *I now invite **Abdullah Zafar Khan** to present **Module 03: Campaigns & Asset Registry**."*

🎙️ PART 3: Abdullah Zafar Khan (Seat No: 2312467) — Campaigns & ERD `6:30 - 10:00 (3 Min 30 Sec)`

Visual Action: Advance to Slide 7 (Module 03), navigate to `/campaigns` .

> *"Thank you, Mudasir. Respected committee members, my name is **Abdullah Zafar Khan**, and I architected **Module 03: Campaign Operations, Geographical Hierarchy & Asset Registry**.*
>
> *Out-of-home market research takes place across vast physical terrain. To model this, I engineered a **5-Tier Pakistan Geographical Location Tree**:*
>
> • **Country:** Pakistan | **Province:** Punjab, Sindh, KPK, Islamabad | **City:** Islamabad, Rawalpindi, Lahore, Karachi
> • **Area:** Blue Area, F-6 Markaz, Gulberg III, Clifton Block 2 | **Site:** Specific geocoded road intersections like *'Centaurus Flyover LED Screen'*.
>
> *Under the **Billboard Media Registry**, we catalog physical dimensions (e.g. `60x20 ft`), illumination tech (Frontlit, Backlit, SMD Digital LED), facing orientation (North-Facing), and high-resolution photo references."*

Visual Action: Click Competitor Intelligence, then open Campaign Evidence tab.

> *"We also built **Competitor Intelligence**: MARP provides comparative side-by-side benchmarking — auditing Jazz's share of voice against Zong 4G and Telenor across key transit corridors.*
>
> *Now, how do we enforce ground truth?*
>
> 1. **Field Staff Dispatch:** Managers schedule surveyors to inspect specific billboard sites with attached questionnaires.
> 2. **Photographic Proof:** The surveyor visits the physical structure and submits evidence.
> 3. **Anti-Fraud Geofencing:** In our Evidence Review workspace, submission #4 was **automatically rejected** because our backend calculated the surveyor's geotag was taken **340 meters outside the authorized geofence radius**! Fraud is caught instantly.

Visual Action: Advance to Slide 4 (Master 3-Circle ERD), or switch browser to `/erd` .

> *"To unify these three modules, our relational schema consists of **28 fully normalized tables**, partitioned into **three distinct student responsibility circles**:*
>
> • **Circle 1 (Indigo — Aroon Kumar):** 10 Entities governing Companies, Users, Roles, Permissions, Devices, and Audit Logs.
> • **Circle 2 (Emerald — Mudasir Ali):** 8 Entities managing Questionnaires, Versions, Questions, Options, Translations, and Branching Rules.
> • **Circle 3 (Amber — Abdullah Zafar Khan):** 10 Entities orchestrating Campaigns, 5-Tier Locations, Advertisements, Competitors, Assignments, and Evidence.
>
> *Foreign key conduits connect our circles: Company tenancy anchors all tables; authorized staff from Circle 1 receive assignments in Circle 3; and certified questionnaire versions from Circle 2 are deployed to campaigns in Circle 3.*
>
> *In conclusion, our platform is fully functional, deployed live on Vercel and Railway, backed by PostgreSQL, and verified with 100% automated test coverage. On behalf of Aroon, Mudasir, and myself — thank you for your time and guidance. We welcome your questions."*

💡 Anticipated Evaluator Technical Questions & Verified Model Answers

`Post-Defense Q&A Defense Guide`

Q1: "How do you prevent surveyors from spoofing GPS using mock location apps?"

AROON / ABDULLAH: "Our mobile client registration captures hardware identifiers and device properties. On modern Android clients, the Google Play Services Location API exposes `location.isFromMockProvider()`. When mock locations are active, the API flags the payload immediately. In addition, our backend computes the Haversine distance between the photo EXIF GPS and the registered billboard coordinate, automatically rejecting any submission outside the geofence tolerance."

Q2: "What happens if a research manager edits a questionnaire while field staff are actively taking surveys?"

MUDASIR: "We engineered an immutable version snapshot pattern. When a questionnaire is published, its exact JSON question schema is frozen as Version 1.0. Any subsequent edits automatically fork into a new draft (v2.0). Field staff continue executing against the certified Version 1.0 snapshot. Historical survey responses are never corrupted or invalidated."

Q3: "Can a field surveyor modify campaign budgets, delete surveys, or approve their own evidence?"

AROON: "No. Role-based access control is enforced on the FastAPI backend using `require_role(...)` dependency injection. Endpoints like evidence approval or campaign deletion strictly require 'Evidence Reviewer' or 'Company Administrator' roles. If a Field Staff bearer token attempts a privileged request, FastAPI returns an immediate HTTP 403 Forbidden with an immutable audit entry."

Q4: "Why did you choose PostgreSQL over MongoDB for MARP?"

ABDULLAH: "MARP is an enterprise audit platform requiring strict ACID compliance, referential integrity, and cascading foreign key constraints across 28 interrelated entities (e.g., deleting a campaign cascades to assignments, while preventing dangling evidence records). In MongoDB, foreign key consistency, multi-table joins, and multi-tenant scoping must be handled manually in application code, introducing risk of orphan records and audit corruption."

Q5: "How does the Urdu localization work under the hood?"

MUDASIR: "In our database, the `question_translations` table stores translated text keyed by `question_id` and `language_code='ur'`. On the frontend, Next.js dynamically swaps the reading direction to `dir='rtl'`, applying custom CSS font classes for Nastaliq calligraphy alignment without reloading the page or losing current responses."